QUADRATIC ITERATOR · ARCHIVED RESULT · GRANULARITY SERIES II

# Iterator Granularity Sweep · Within-Domain

Overfit scaling vs encoding resolution · Cross-domain
Train: Genesis Ch.1–10 · Test: Alice Ch.I

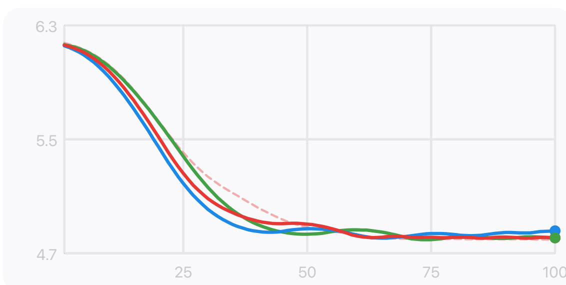
D=3 fixed input | CTX=32 | 128×2 MLP | 100 epochs | lr=0.003 | Vw=482 | seed=42

## LOSS CURVES · TRAIN (DASHED) VS TEST (SOLID)

Iterator (C) test   - - - Iterator train   Raw (A) test   Random (R) test

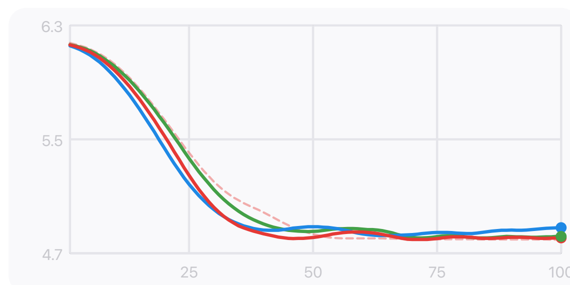
### W=1

1 bit per token · 1-bit



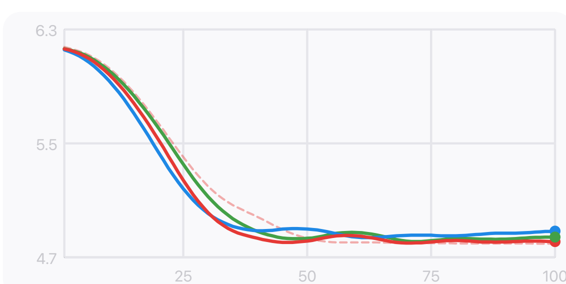

### W=8

8 bits per token · character



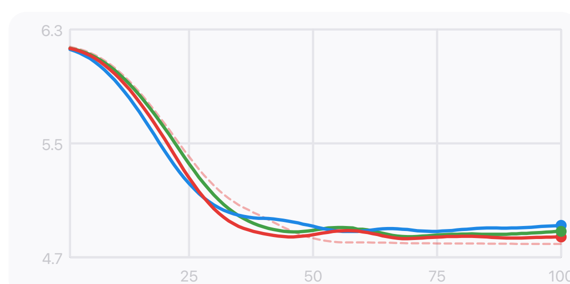

### W=16

16 bits per token · bigram




### W=24

24 bits per token · triplet




## GENERALISATION GAP (TEST − TRAIN) AT EPOCH 100

| RESOLUTION | C GAP | R GAP | A GAP | C − R |
|---|---|---|---|---|
| **W=1**<br>1-bit | +0.0130 | +0.0099 | +0.3133 | +0.0031 |
| **W=8**<br>character | +0.0126 | +0.0390 | +0.3032 | -0.0264 |
| | | | | |

| | | | | |
|---|---|---|---|---|
| **W=16**<br>bigram | +0.0151 | +0.0833 | +0.2772 | -0.0682 |
| **W=24**<br>triplet | +0.0496 | +0.1475 | +0.3416 | -0.0979 |

KEY FINDING

### C sits on the parity axis

Iterator overfit range: +0.013 to +0.050 across 24× resolution change — near zero throughout

C − R TREND

### R tracks C at fine windows

At W=1, C−R≈+0.003 (indistinguishable). Gap opens as window grows to −0.098 at W=24

## CONDITIONS

| LABEL | DESCRIPTION |
|---|---|
| C · Iterator | [mean_b0, mean_b1, D*_digest] · quad averaged over W bits · digest at word boundary · fixed input |
| A · Raw | Raw triplet tokens · learned D=3 embedding · trained end-to-end |
| R · Random | Gaussian noise D=3 · fixed input · control condition |